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MATHEMATICS

STANDARD SIX

Midterm exam - August 2024
Answer Key and Worked Solutions

Subject	Mathematics
Code	Not printed in source
Level	Standard Six
Year	2024
Exam Board / Issuer	MESP Tanzania
Exam	Midterm exam - August
Type	Answer key and worked solutions
Questions	33 numbered questions/subparts

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: Mathematics-S6-2024-MESP-Tanzania-August-Midterm-exam.pdf

Notes about the source paper

Items with missing information or printing errors are explained in their solutions. A model response is an example, rather than a prescribed official answer.

Calculations

1(i). Add $47856 + 9918$.

Step 1: 1 place: $6 + 8 = 14$; write 4, carry 1.

Step 2: 10 place: $5 + 1 + \text{carry } 1 = 7$; write 7.

Step 3: 100 place: $8 + 9 = 17$; write 7, carry 1.

Step 4: 1000 place: $7 + 9 + \text{carry } 1 = 17$; write 7, carry 1.

Step 5: 10000 place: $4 + 0 + \text{carry } 1 = 5$; write 5.

Step 6: $47856 + 9918 = 57774$.

Final Answer: 57774

1(ii). Subtract $52013 - 7892$.

Step 1: 1 place: $3 - 2 = 1$.

Step 2: 10 place: regroup 1 from the next place; $11 - 9 = 2$.

Step 3: 100 place: regroup 1 from the next place; $9 - 8 = 1$.

Step 4: 1000 place: regroup 1 from the next place; $11 - 7 = 4$.

Step 5: 10000 place: $4 - 0 = 4$.

Step 6: $52013 - 7892 = 44121$; check $44121 + 7892 = 52013$.

Final Answer: 44121

1(iii). Multiply 568×47 .

Step 1: Split into place values: $500 \times 47 = 23500$; $60 \times 47 = 2820$; $8 \times 47 = 376$.

Step 2: Add the partial products: 26696.

Final Answer: 26696

1(iv). Divide 26696 by 47.

Step 1: $266 \div 47$: write 5; $5 \times 47 = 235$; $266 - 235 = 31$.

Step 2: $319 \div 47$: write 6; $6 \times 47 = 282$; $319 - 282 = 37$.

Step 3: $376 \div 47$: write 8; $8 \times 47 = 376$; $376 - 376 = 0$.

Step 4: $47 \times 568 = 26696$.

Step 5: $26696 \div 47 = 568$.

Final Answer: 568

Section Summary: Read each value and unit carefully. Align place values and check calculations by reversing the operation. For pictures, count only the requested objects.

Mathematics

1(ix). Add 73.92 + 10.3.

Step 1: Write 10.30 aligned.

Step 2: Hundredths $2+0=2$; tenths $9+3=12$, write 2 carry 1.

Step 3: Whole $73+10+1=84$.

Final Answer: 84.22

1(v). Add $12 \frac{1}{4} + 5 \frac{3}{4}$.

Step 1: Whole $12+5=17$; fractions $\frac{1}{4}+\frac{3}{4}=1$.

Step 2: $17 + 1 = 18$.

Final Answer: 18

1(vi). Subtract $5 \frac{2}{3} - 1 \frac{3}{5}$.

Step 1: Convert $\frac{17}{3}$ and $\frac{8}{5}$.

Step 2: Common denominator 15: $\frac{85}{15} - \frac{24}{15}$.

Step 3: $85 - 24 = 61$; $\frac{61}{15} = 4 \frac{1}{15}$.

Final Answer: $4 \frac{1}{15}$

1(vii). Multiply $3\frac{1}{4} \times 3\frac{1}{5}$.

Step 1: Convert $13/4$ and $16/5$.

Step 2: Cancel $16/4=4$.

Step 3: $13 \times 4 = 52$; denominator 5.

Step 4: $52/5=10\frac{2}{5}$.

Final Answer: $10\frac{2}{5}$

1(viii). Divide $3\frac{3}{4}$ by $2\frac{1}{2}$.

Step 1: Convert $15/4$ and $5/2$.

Step 2: Multiply $15/4$ by $2/5$.

Step 3: $30/20=3/2=1\frac{1}{2}$.

Final Answer: $1\frac{1}{2}$

1(x). Subtract $11.2 - 8.945$.

Step 1: Align 11.200 and 8.945.

Step 2: Borrow:10 whole,11 tenths,9 hundredths,10 thousandths.

Step 3: $10-5=5$; $9-4=5$; $11-9=2$; $10-8=2$.

Final Answer: 2.255

2(i). Write 346581 in words.

Step 1: $346581=346000+581$.

Final Answer: Three hundred and forty-six thousand five hundred and eighty-one

2(ii). Continue 1,6,4,9,7,[].

Step 1: Alternate $+5, -2$.

Step 2: $7 + 5 = 12$.

Final Answer: 12

2(iii). Arrange -10, +2, -1, -3, 0, +3 in ascending order.

Step 1: Negative numbers precede 0; more negative is smaller.

Final Answer: -10, -3, -1, 0, +2, +3

2(iv). Name each place in 7693251.

Step 1: Read seven places left to right.

Final Answer: 7 millions; 6 hundred-thousands; 9 ten-thousands; 3 thousands; 2 hundreds; 5 tens; 1 ones.

2(v). Write Ashiruna age 35 in Roman numerals.

Step 1: $30 = XXX$; $5 = V$.

Final Answer: XXXV

2(vi). Mean masses 200, 400, 600, 400, 700 kg?

Step 1: $200 + 400 + 600 + 400 + 700 = 2300$.

Step 2: $2300 / 5 = 460$.

Final Answer: 460 kg

3(i). Najati leaves 18:30 and travels 30 minutes. Arrival 24-hour time?

Step 1: $18:30 + 30 \text{ minutes} = 19:00$.

Final Answer: 19:00

3(ii). Shoes bought 25000 sh, sold 16000 sh. Loss?

Step 1: $\text{Loss} = \text{buying} - \text{selling}$.

Step 2: $25000 - 16000 = 9000$.

Final Answer: Sh 9000

3(iii). 28 teachers,18 female. Fraction male?

Step 1: $28 - 18 = 10$ males.

Step 2: $10/28=5/14$.

Final Answer: 5/14

4(i). Convert 21.45%to decimal.

Step 1: Divide percentage by 100.

Step 2: $21.45 / 100 = 0.2145$.

Final Answer: 0.2145

4(ii). Solve $4y + 6 = 30$.

Step 1: $4y = 30 - 6 = 24$.

Step 2: $24 / 4 = 6$.

Step 3: Check $4 \times 6 + 6 = 30$.

Final Answer: $y=6$

4(iii). Machine costs 35000 shand sells for 5%profit. Selling price?

Step 1: $35000 \times 5 / 100 = 1750$ profit.

Step 2: $35000 + 1750 = 36750$.

Final Answer: Sh 36750

5(i). Add 5 kg 650 gand 4 kg 500 g.

Step 1: $650 + 500 = 1150$ g= 1 kg 150 g.

Step 2: $5 + 4 + 1 = 10$ kg.

Final Answer: 10 kg 150 g

5(ii). Number below 10 which is both even and prime?

Step 1: Only 2 has exactly 2 factors andis even.

Final Answer: 2

5(iii). Add 45%to 0.25; give percentage.

Step 1: $0.25=25\%$.

Step 2: $45 + 25 = 70$.

Final Answer: 70%

6(i). Find the GCF of 48,60,96.

Step 1: $48=2^4 \times 3$; $60=2^2 \times 3 \times 5$; $96=2^5 \times 3$.

Step 2: Shared lowest powers $2^2 \times 3=12$.

Final Answer: 12

6(ii). List odds between 26 and 46.

Step 1: Start 27 and add 2 until 45.

Final Answer: 27,29,31,33,35,37,39,41,43,45

6(iii). Simplify $n + 2b + 3n - b$.

Step 1: ncoefficients $1+3=4$; bcoefficients $2-1=1$.

Final Answer: $4n + b$

7(i). Graph sales Monday 200, Tuesday 600, Wednesday 200, Thursday 700, Friday 400. Wednesday+Friday?

Step 1: $200 + 400 = 600$.

Final Answer: 600 chickens

7(ii). Mean chicken sales 200,600,200,700,400 in 5 days?

Step 1: $200 + 600 + 200 + 700 + 400 = 2100$.

Step 2: $2100 / 5 = 420$.

Final Answer: 420 chickens

8(i). Rectangle length 40 cm, perimeter 120 cm. Find area.

Step 1: Length+width $120 / 2 = 60$.

Step 2: Width $60 - 40 = 20$.

Step 3: $40 \times 20 = 800$.

Final Answer: 800 cm²

8(ii)-circle. Circle diameter 28 cm, pi 22/7. Circumference?

Step 1: Circumference= $\pi \times \text{diameter}$.

Step 2: $(22/7) \times 28 = 88$.

Final Answer: 88 cm

8(ii)-triangle. Right triangle legs 8 cm,6 cm; hypotenusey. Findy.

Step 1: Pythagoras: $y^2=8^2+6^2$.

Step 2: $64 + 36 = 100$.

Step 3: $10 \times 10 = 100$, so $y=10$.

Final Answer: 10 cm

Section Summary: Read each value and unit carefully. Align place values and check calculations by reversing the operation. For pictures, count only the requested objects.

Source Checks

Darasa Huru - Standard Six Midterm Exams August 2024 By MESP Tanzania